Problem 1.11

You have \$10,000 to invest now and are being offered \$22,500 after ten years as the return from the investment. The market rate is 10% compound interest. Ignoring complications such as the effect of taxation, the reliability of the company offering the contract, etc., do you accept the investment?

Solution

To decide whether to accept the investment, compare the offered future value with the future value obtained by investing \$10,000 at the market rate of 10% compounded annually for 10 years.

Using the compound interest formula,

$$A = P(1+i)^t,$$

with

$$P = 10{,}000, \qquad i = 0.10, \qquad t = 10,$$

we get

$$A = 10{,}000(1.1)^{10}.$$

Calculating,

$$A \approx 10{,}000(2.59374246) = 25{,}937.42.$$

The offered amount is

$$22{,}500,$$

which is less than

$$25{,}937.42.$$

The difference is

$$25{,}937.42 - 22{,}500 = 3{,}437.42.$$

Therefore, the investment should **not** be accepted, because it pays about \$3,437.42 less than what could be earned at the market rate.